

TP 3 : Base de données Cassandra

```
C:\Windows\System32>docker pull cassandra
Using default tag: latest
latest: Pulling from library/cassandra
7a2c55901189: Pull complete
5aa209e79be1: Pull complete
fb3b10c5302c: Pull complete
b138106695bf: Pull complete
3c6299c5a721: Pull complete
43a5eb41ded0: Pull complete
ffdf1e666a30: Pull complete
b3a102ae504c: Pull complete
ef0b671ac11f: Pull complete
3615cc882ec3: Pull complete
Digest: sha256:e7410759812c49ebb71740e40237158f115b48e3ad9d956bb4e4224c68f15649
Status: Downloaded newer image for cassandra:latest
docker.io/library/cassandra:latest

What's Next?
  View a summary of image vulnerabilities and recommendations → docker scout quickscan

C:\Windows\System32>docker images
REPOSITORY    TAG       IMAGE ID      CREATED        SIZE
cassandra     latest   60f97b970450  4 months ago  355MB

C:\Windows\System32>
```

```
cqlsh 127.0.0.1 9042
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.3 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE mykeyspace WITH REPLICATION = { 'class' : 'SimpleStrategy', 'replication_factor' : 1 };
cqlsh> describe keyspaces;

mykeyspace  system_auth          system_schema  system_views
system      system_distributed    system_traces  system_virtual_schema

cqlsh> USE mykeyspace;
cqlsh:mykeyspace>
```

```

cqlsh:mykeyspace> CREATE INDEX ON users (lname);
cqlsh:mykeyspace> SELECT * FROM users WHERE lname = 'smith';

  user_id | fname | lname
-----+-----+-----
    1745 |  john |  smith

(1 rows)
cqlsh:mykeyspace> CREATE TABLE tab2
... (id1 int,id2
... int,first_name
... varchar,last_name
... varchar, PRIMARY
... KEY (id1, id2));
cqlsh:mykeyspace> ALTER TABLE users ADD telephone text;

```

```

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```

```

cqlsh:mykeyspace> UPDATE users SET telephone = '21212121' where user_id=1745
;
cqlsh:mykeyspace> TRUNCATE users;
cqlsh:mykeyspace> DROP table mykeyspace.users;

```

```

cqlsh:mykeyspace> CREATE TABLE users1(
... user_id int PRIMARY KEY,
... fname text,
... lname text,
... emails set<text>
... );
cqlsh:mykeyspace> INSERT INTO users1 (user_id, fname, lname,
... emails) VALUES(1234, 'Frodo', 'Baggins',
... {'f@baggins.com', 'baggins@gmail.com'});
cqlsh:mykeyspace> UPDATE users1
... SET emails = emails + {'fb@friendsofmordor.org'}
... WHERE user_id = 1234;
cqlsh:mykeyspace> SELECT user_id, emails FROM users1 WHERE user_id =
... 1234;

  user_id | emails
-----+-----
    1234 | {'baggins@gmail.com', 'f@baggins.com', 'fb@friendsofmordor.org'}

(1 rows)

```

```

cqlsh:mykeyspace> UPDATE users1 SET emails = emails - {'fb@friendsofmordor.
rg'}
... WHERE user_id = 1234;
cqlsh:mykeyspace> UPDATE users1 SET emails = {} WHERE user_id =
... 1234;
cqlsh:mykeyspace> ALTER TABLE users1 ADD top_places list<text>;
cqlsh:mykeyspace> UPDATE users1
... SET top_places = [ 'rivendell', 'rohan' ]
... WHERE user_id = 1234;
cqlsh:mykeyspace> UPDATE users1
... SET top_places = top_places + [ 'mordor' ]
... WHERE user_id = 1234;
cqlsh:mykeyspace> UPDATE users1
... SET top_places[1] = 'riddermark' WHERE user_id = 1234;
cqlsh:mykeyspace> DELETE top_places[2] FROM users1 WHERE user_id = 1234;
cqlsh:mykeyspace> SET top_places = top_places - ['riddermark']
... WHERE user_id = 1234;

```

```

cqlsh:mykeyspace> SELECT user_id, top_places FROM users1 WHERE user_id = 123
4;

user_id | top_places
-----+-----
    1234 | ['rivendell', 'riddermark']

(1 rows)
cqlsh:mykeyspace> ALTER TABLE users1 ADD todo map<timestamp, text> ;
cqlsh:mykeyspace> UPDATE users1 SET todo = {
... '2012-09-24' : 'enter mordor',
... '2012-10-2 12:00' : 'throw ring into mount doom'
... } WHERE user_id = 1234;
cqlsh:mykeyspace> UPDATE users1
... SET todo['2012-10-2 12:00'] = 'throw my precious'WHERE
... user_id = 1234;
cqlsh:mykeyspace> INSERT INTO users1 (user_id,todo) VALUES ( 1234,
... { '2013-09-22 12:01' : 'birthday wishes to Bilbo',
... '2013-10-1 18:00' : 'Check into Inn of Prancing Pony'})
... });

```

```

C:\Users\ghada>pip install cassandra-driver
Collecting cassandra-driver
  Downloading cassandra-driver-3.28.0.tar.gz (291 kB)
    291.6/291.6 kB 562.7 kB/s eta 0:00:00
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
Collecting six>=1.9 (from cassandra-driver)
  Using cached six-1.16.0-py2.py3-none-any.whl (11 kB)
Collecting geomet<0.3,>=0.1 (from cassandra-driver)
  Downloading geomet-0.2.1.post1-py3-none-any.whl (18 kB)
Collecting click (from geomet<0.3,>=0.1->cassandra-driver)
  Obtaining dependency information for click from https://files.pythonhosted.org/packages/00/2e/d53fa4befbf2cfa713304aff
e7ca780ce4fc1fd8710527771b58311a3229/click-8.1.7-py3-none-any.whl.metadata
  Downloading click-8.1.7-py3-none-any.whl.metadata (3.0 kB)
Collecting colorama (from click->geomet<0.3,>=0.1->cassandra-driver)
  Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
  Downloading click-8.1.7-py3-none-any.whl (97 kB)
    97.9/97.9 kB 701.6 kB/s eta 0:00:00
Building wheels for collected packages: cassandra-driver
  Building wheel for cassandra-driver (pyproject.toml) ... done
  Created wheel for cassandra-driver: filename=cassandra_driver-3.28.0-cp312-cp312-win_amd64.whl size=335612 sha256=8d56
fa57decfcfa8b09a082cfe8ddd116fde1498ccee8f981bd1883587882dab
  Stored in directory: c:\users\ghada\appdata\local\pip\cache\wheels\50\8d\67\2affaba40f98e1ffa772fa9178ce15812c6642e69
604534

```

IX. Manipulations de Cassandra avec Python

4 e. Exprimer en CQL les requêtes suivantes :

1. Liste de tous les restaurants.

```
SELECT * FROM restaurants;
```

2. Liste des noms de restaurants.

```
SELECT name FROM restaurants;
```

3. Nom et quartier (borough) du restaurant N° 41569764.

```
SELECT name, borough FROM restaurants WHERE id = 41569764;
```

4. Dates et grades des inspections de ce restaurant.

```
SELECT inspection_date, grade FROM inspections WHERE restaurant_id = 41569764;
```

5. Noms des restaurants de cuisine Française (French)

```
SELECT name FROM restaurants WHERE cuisine_type = 'French';
```